

CSA-1610

S. Md. Irfan

192373001

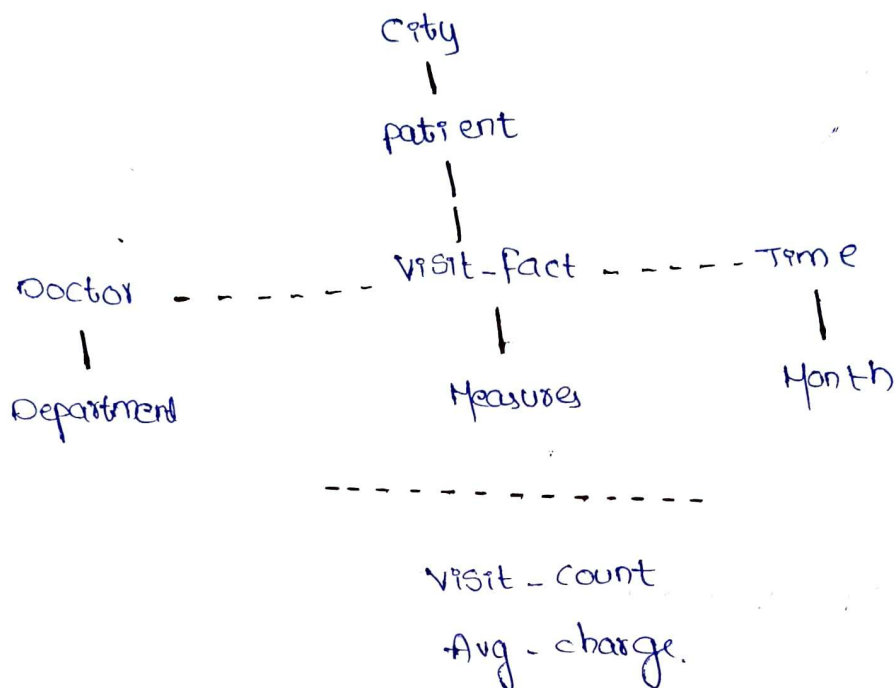
C.S-E(D.S)

COI - AT1

Q1)
sol

City Hospital Data warehouse

(a) Snowflake Schema.



Fact Table : visit - fact

- * patient - ID
- * doctor - ID
- * Department - ID
- * Time - ID
- * visit - count
- * Avg - charge.

Dimension Table :

patients

- * patient - ID
- * Name.

- * Age
- * Gender
- * City

Doctor

- * Doctor-ID
- * Name
- * Specialization

Department

- Department-ID
- Department-Name

Time

- * Date
- * Month
- * Quarter
- * year.

b) OLAP Operations

starting cuboid : [patient, Doctor, Department, Time]

Operations:

1. Roll-up patient $\rightarrow$ ALL
2. Roll-up Doctor $\rightarrow$ ALL
3. Roll-up Time $\rightarrow$ year
4. keep department unchanged.
5. Calculate Avg (Avg-change)

Final Cuboid :

[ALL, ALL, Department, year]

c) Number of cuboids

Dimensions = 4

level per dimension = 4

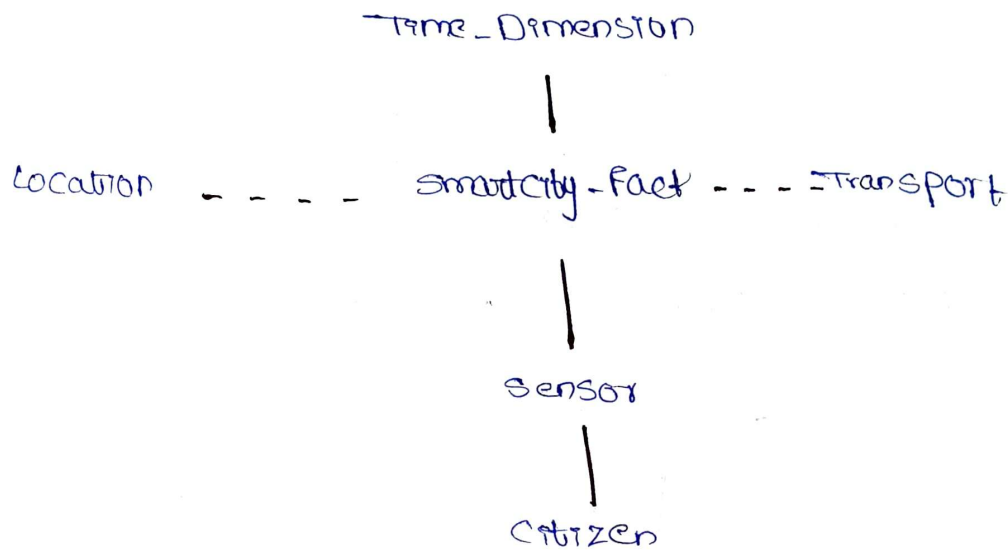
Formula $\Rightarrow$ cuboids = $4 \times 4 \times 4 \times 4 = 4^4 = 256$ cuboids.

Q2)

sol

Smart City Star Schema.

Star schema.



Fact Table

Smart-city Fact

Measures :

- $\rightarrow$ Vehicle count
- $\rightarrow$ congestion level
- $\rightarrow$ Energy consumption
- $\rightarrow$ Air quality index
- $\rightarrow$ Noise level
- $\rightarrow$ Ridership
- $\rightarrow$ Punctuality
- $\rightarrow$ citizen feedback score.

Dimension Tables

Time

- * Day
- * Month
- * Year

Location

- * City
- * Zone
- * Road

Sensor

- * Sensor ID
- * Sensor Type

Transport

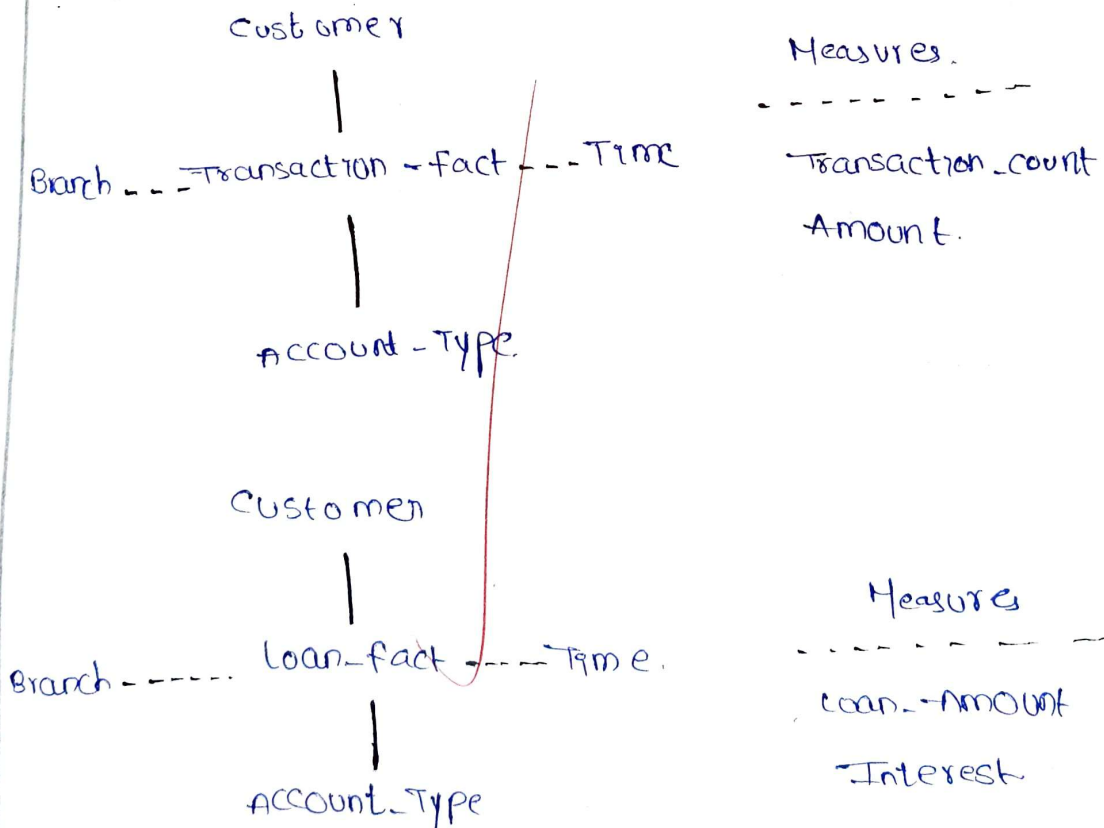
- * Bus
- * Metro
- * Route

Citizen

- * User ID
- * Age group
- * Feedback category

Q3) Banking Data Warehouse.

a) Galaxy Schema.



Shared Dimensions

- customer
- Branch
- Time
- Account type.

b) OLAP operations

Base cuboid : [customer, Branch, Time - Account - Type]

operations

- * Roll-up customer → ALL
- * Roll-up Account-Type → ALL
- * Roll-up Time → year
- * Keep Branch

Measures

- Avg(Amount)
- Avg(loan-Amount)

Final cuboid : [ALL, Branch, year, ALL]

c) Cuboids

Dimension = 4 , level = 4

Total cuboids = $4^4 = 256$ cuboids

Q4) Manufacturing company

Roll up : Summarize defect rates by product category for the current year across all factories

operation

- product → category
- factory → ALL
- Time → current year

Measure

- * Defect rate

Slice

- condition : supplier rating ≥ 4
- Measure : Production units
- Across : all regions

Drill down

- * factory → Department
- * product → product type

Measure

- Defect rate

Drice

- * filters : selected product
- * selected factory location

Measures :

- * Defect rate
- * Supplier reliability.

Q5) Retail chain

Roll up

Summarize Inventory Turnover by Product Category.

Drill Down

Filter :

- * product category = Electronics
- * Region = Selected region

Drill down to

- * Sub category
- * SKU

Measures

- * Sales revenue
- * Units sold

Slice

Filter

Gender = Male

Membership = Bronze

Measures : Sales revenue

ACROSS : All stores

Dice

Filters

- * selected cities
- * selected product categories.
- * Selected time periods.

Measures

- sales revenue
- units sold.